COMPARATIVE TABLE OF JAVA,C,C++,PYTHON AND JAVASCRIPT.

parameters	JAVA	C	C++	PYTHON	Javascript
Language / Scripting	Language	Language	Language	Mixed	Scripting
Open Source or Commercial	Open source &commercia l	Open source	Open source	Open source	Open source
Compiler or Interpreter	Complier	Complier	Complier	Interpreter	Mixed
Object-Oriented Programming Support (Yes/No)	Yes	No	Yes	Yes	Yes
Developer Organization	Sun microsystem	ISO/IEC WG14	ISO/IEC WG21	Python Software Foundation(PSF)	Ecma International
Developer (Person)	James Gosling	Dennis Retchie	Bjame Stroustrup	Guido van Rossum	Brendan Eich
Current Major Version	Java26	C23	C++23	Python 3.14	ES2025
Primary Purpose	Large-scale business software	System-level software	Performance-critical software	Artificial Intelligence, machine learning	Web browser interactivity
Common Applications	Spotify, games, Netflix	Smart home frim ware, ios	Game engines, Video process	Instagram , websites	Frontend Airbnb,web applications
Database	YES	YES	YES	YES	YES
Memory Management	Automatic	Manual	Manual	Automatic	Automatic
Security	YES	NO	NO	YES	YES
Performance	Moderate	Very high	High	Very low	Low
Platform Independence	YES	NO	NO	YES	YES
Other Important Features	Write once, run Anywhere	Interactive with hardware	Highly extensible	Massive ecosystem	Asynchronous execution
Ease of Learning	Moderate	Easy	Moderate to Difficult	Very Easy	Moderate
Popular IDEs	IntelliJ Idea	C Lion	C Lion	PyCharm	Web Strom

Compilation Output	Compiled to byte code	Compiled	Compiled	Interpreter	Interpreted /JIT Compiled
Advantages	Universal platform cross-compatibility	Maximum execution speed, direct interaction with hardware	Multi-paradigm structural support	Ultra-clean syntax	Unrivaled browser dominance
Limitations	Heavy memory usage	No object-oriented design	Severe security flaws	Slow processing throughput	Client-side exposure risk